A Centralized System for Real-Time Farm Animal Disease Reporting

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Reactive veterinary healthcare model in livestock industry of Sri Lanka contributes to a significant economic loss since the response usually comes when the animals have exhibited serious clinical signs or when the animals are dead. To resolve this problem, the proposed solution is the "ADRS: Automated Cattle Wellness Monitoring System which proposes a 4-level model proactive system that involves wellness triage, disease detection, spread prediction and seasonal prognosis. The most vital level of this study is AI-Driven Wellness and Management Standardization. This component operates via a Dual-Trigger Diagnostic model in contrast to the Precision Livestock Farming (PLF) systems which are expensive and need 3D cameras and sensors.

The system first serves as a continuous tracker, with the farmers recording cattle with initial parameters (age, initial weight, gender, breed) and inputting daily milk production and body weight scale-data. The 7-day Deep Diagnostic triage window may be triggered either automatically, when the system notices statistical abnormalities in the day-to-day logs, or externally by the farmer in case he notices the appearance of unusual behavior. In this window, the system matches the daily feed, milk, and weight records with environmental data collected by GPS in determining the Temperature-Humidity Index (THI). The system combines these multi-modal features with a final 2D mobile image to predict the animal as a Healthy, at risk, or unhealthy animal using a YOLOv8 and CNN-based Computer Vision pipeline. Grad-CAM heatmaps provide transparency, which allows early intervention to ensure that the seasonal milk production is maximized and the national milk supply chain is stabilized.

Keywords: Proactive Triage, Dual-Trigger Diagnostics, Body Condition Score (BCS), Multi-modal Fusion, Heat Stress (THI), Computer Vision, Anomaly Detection, Explainable AI (XAI).

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LIST OF ABBREVIATIONS

- **ADRS:** Automated Cattle Wellness Monitoring System
- **AI:** Artificial Intelligence
- **API:** Application Programming Interface
- **BCS:** Body Condition Score / Scoring
- **CI/CD:** Continuous Integration and Continuous Deployment
- **CLAHE:** Contrast-Limited Adaptive Histogram Equalization
- **CNN:** Convolutional Neural Network
- **DSR:** Design Science Research
- **GPS:** Global Positioning System
- **Grad-CAM:** Gradient-weighted Class Activation Mapping
- **IoU:** Intersection over Union
- **IoT:** Internet of Things
- **JSON:** JavaScript Object Notation
- **JWT:** JSON Web Token
- **mAP:** Mean Average Precision
- **PLF:** Precision Livestock Farming
- **RMSE:** Root Mean Square Error
- **SDG:** Sustainable Development Goals
- **THI:** Temperature-Humidity Index
- **UML:** Unified Modeling Language
- **XAI:** Explainable Artificial Intelligence
- **YOLO:** You Only Look Once

1. INTRODUCTION

1.1 Background and Context

1.1.1 Overview of the Global and Local Dairy Industry

Agricultural sector, especially livestock and dairy farming, is one of foundations of food security and economic stability of the world. The dairy industry is a crucial sector of Sri Lankan economy, it directly employs more than 300,000 rural families and it is the main source of vital nutrients [1]. Even with its government efforts, it is estimated that Sri Lanka can now only satisfy approximately 40 percent of its domestic dairy needs with the heavy reliance on imported milk products that cost the government billions of rupees of income per year [2]. To close this gap, there has been a shift by most of the small-to-medium-scale farmers to having high yielding foreign cattle breeds like the Jersey and Holstein-Friesian. Nevertheless, the physiological challenge of keeping such breeds that are sensitive to climate in a tropical environment is very harsh [3].

1.1.2 The Core Problem: Reactive Healthcare and Economic Loss

The inherent problem that is affecting the dairy industry in Sri Lanka negatively is the high dependency on a reactive veterinary healthcare model [4]. Farmers do not normally consult medical attention until an animal has shown serious, observable clinical signs, including absolute lethargy, abrupt recumbency or a drastic reduction in the level of milk production. The biological damage to metabolic system of the cow at this advanced stage is usually irreversible. In addition, the infectious diseases can have transferred to the remainder of the herd. This wait-and-see strategy leads to **high mortality rates, massive veterinary costs**, and a probable 20-30 percent loss in the possible milk production per lactation cycle [5].

1.1.3 Limitations of Existing Solutions

Although the global trends in the fields of Smart Agriculture and PLF involve the use of AI to proactively monitor the state of animals, the solutions are skewed towards large-scale industrial farms [6].

Manual Subjectivity: In Sri Lanka, the evaluation of a cow on energy balance is carried out by Body Condition Scoring (BCS), a technique that requires the human to touch and feel the cow's bones. This is very subjective and quite different among observers [7].

Technological Unavailability: Current automated PLF solutions need 3D depth cameras and continuous IoT sensors, but the cost of these technologies is prohibitive for small-scale farmers [8].

Neglecting Environmental Stressors: Foreign breeds automatically decrease food consumption and milk production when the Temperature-Humidity Index (THI) exceeds 72. This biological reaction to heat is often mistaken by farmers as a clinical disease, and hence the resultant wrong medication [9].

1.1.4 The Proposed Solution: ADRS Overarching Framework

To mitigate these problems, a 4-tier proactive intelligence model is presented in the "ADRS," intended for use in tropical small-scale farms:

- **Wellness Triage (This Component):** Evaluates the pre-clinical health condition based on multi-modal data fusion.
- **Disease Detection:** Classifies specific pathological diseases based on visual features.
- **Disease Spread Prediction:** Predicts the geographical spread of identified diseases to neighboring farms.
- **Seasonal Forecasting:** A method that uses past records of meteorological and health data to forecast seasonal outbreaks of diseases.

1.1.5 Focus of this Research Component: AI-Driven Wellness Standardization

This study element is dedicated to Tier 1: Wellness Triage, which presents a new model of Dual-Trigger Diagnostic [10]. The system would initially be considered as a continuous monitoring system wherein farmers would enroll cattle with baseline parameters, such as age, weight, gender, and breed before recording daily rates such as milk yield and body weight scale-data [12]. The Deep Diagnostic triage window may be set on the 7 days basis in two different ways: automatically, when the system detects a statistical deviation in these daily production and weight records, or by the farmer himself, when he notices a strange behavior [13]. In this window, the system matches daily feed, milk and weight records with environmental data obtained by GPS and then calculates the Temperature-Humidity Index (THI) [19]. The system uses a YOLOv8 pipeline and CNN-based pipeline which combines all these multi-modal features with a final 2D mobile image to classify the animal as Healthy, At Risk or Unhealthy [20]. The transparency is achieved by the use of Grad-CAM heatmaps, which gives the farmers visual confirmation and interpretability of the final classification offered by the AI [21].

1.1.6 Alignment with the Sustainable Development Goals (SDGs)

The present study corresponds to SDG 2 (Zero Hunger) because it optimizes the production of animal protein and reduces livestock mortality. Moreover, it is aligned with SDG 9 (Industry, Innovation, and Infrastructure) since it will digitalize rural farming methods and introduce the latest AI-based technologies to underserved agricultural communities.

1.2 Literature Review

Recent academic literature shows an increased interest in automated cattle health with the use of different technological solutions. Nevertheless, an analysis of the literature and commercial products shows that there are major limitations for small-scale tropical farming.

1.2.1 Current Research Methodologies

- **Computer Vision in BCS:** Miekley et al. [7] examined several 2D and 3D automated Body Condition Scoring systems and concluded that 3D systems are accurate, but highly sensitive to the farm environment. More recently, Zingaretti et al. [11] demonstrated that 2D images can be used in estimating the BCS by using YOLOv8. Ocholla et al. [8], however, noted that these 2D models usually struggle with environmental noise such as mud, complex backgrounds, and poor lighting.

- **Multimodal Deep Learning:** Qiao et al. [12] demonstrated that the addition of visual data to physiological consumption logs is much more effective for the pre-symptomatic identification of metabolic disorders. On the same note, Ayaz et al. [6] discussed the advantages of smart agriculture based on the IoT. However, these research models depend heavily on the presence of continuous and automated data gathering.

- **Tropical Climate Stress (THI):** Recent studies by Herbut et al. [13] and other tropical veterinary studies [9] have shown the enormous effect of THI on high-yielding foreign breeds. Once the THI reaches above 72, a biological heat response is activated, and the milk yield and behavior change dramatically.

1.2.2 Current Products and Services

In the commercial market today, industrial Precision Livestock Farming (PLF) systems including Afimilk IoT collars and dairy herd management software like DeLaval DelPro dominate the market. Although it is possible to continuously monitor health with these products, they are based on costly wearable sensors, 3D cameras, and high-speed agricultural infrastructure. Currently, mobile-based applications that serve farmers act mainly as manual diaries and do not consider real-time THI or deliver automated visual health diagnostics.

1.2.3 Limitations and the Requirement of Explainable AI (XAI)

The main weakness of both current research models and commercial products is the fact that they are "Black-box" in nature, and therefore they cannot be easily trusted by rural farmers. In order to fill this gap, recent research [14] proposes applying Gradient-weighted Class Activation Mapping (Grad-CAM) to generate visual heatmaps, thus rendering AI decisions clear and understandable to skeptical users. Moreover, existing solutions create a Triage Gap by not offering an on-demand, mobile-friendly diagnostic platform to small-scale farmers facing urgent pre-clinical issues.

1.3 Research Gap

The review of the literature and available commercial products shows that there are three significant gaps that necessitate this research:

- **The Dimensionality Gap:** The existing high-accuracy BCS solutions demand high-cost 3D depth cameras and permanent IoT infrastructure [8], which are neither economically nor technically accessible to small-scale farmers. The existing strategies are insufficient since they do not leverage available mobile technology. Hence, **there is a missing link for** a robust, low-cost 2D mobile-based vision system that uses advanced preprocessing (CLAHE) to achieve similar diagnostic accuracy.

- **The Tropical Breed Gap:** Existing AI systems and datasets around the world are mostly trained on temperate breeds in well-structured settings. These references critically fail to generalize to the unique morphological adaptations of foreign breeds managed under Sri Lankan tropical conditions [3].

- **The Contextual Triage Gap (Primary Novelty):** Current automated PLF products are exclusively configured to work in a 24/7 continuous mode [12], leaving rural farmers without an affordable alternative. **What is missing** is a proactive, On-Demand Diagnostic Triage Tool. This study presents a highly innovative and original method, providing farmers the opportunity to manually enter a specific 7-day diagnostic window (water, feed, milk) and mathematically correlate it with a 2D image captured via a mobile device and real-time environmental THI. This novel multi-modal fusion forms a highly accessible, context-aware early warning system that is completely unavailable in existing PLF solutions.

Table 1 Comparison with existing applications

Feature / Capability	Existing Commercial Products (e.g., Afimilk, DeLaval)	Previous Academic Research 1 (Zingaretti et al. [11])	Previous Academic Research 2 (Qiao et al. [12])	**Our Proposed System (ADRS - Tier 1)**
Hardware Required	Expensive IoT Sensors / 3D Depth Cameras	Standard 2D Camera	Continuous IoT Sensors	**Low-cost Mobile 2D Camera**
Monitoring Mode	24/7 Continuous Monitoring	Point-in-time assessment	24/7 Continuous Monitoring	**On-Demand 7-Day Triage Window**
Environmental Context	Ignored (or requires separate expensive modules)	Ignored	Ignored	**Integrated (Real-time GPS-based THI)**
Explainable AI (XAI)	No (Black-box logic)	No	No	**Yes (Grad-CAM Visual Heatmaps)**
Suitability for Tropical Small-scale Farmers	Very Low (Costly & requires high-tech infrastructure)	Moderate (Struggles with environmental noise)	Low (Requires continuous data)	**Very High (Mobile-friendly & Context-aware)**

2. OBJECTIVE

2.1 Main Objective

To create an AI-based multi-modal diagnostic triage framework that standardizes livestock wellness monitoring by combining 2D computer vision, 7-day physiological sequences, and real-time environmental data for small-scale tropical dairy farms.

2.2 Specific Objectives

1. To create a Lightweight 2D Computer Vision Framework using YOLOv8 and CNN-Regression for objective BCS estimation on low-end mobile devices.
2. To define Tropical Wellness Benchmarks specifically for livestock breeds that are reared under Sri Lankan climatic conditions.
3. To develop a Persistent Monitoring and Dual-Trigger Fusion model that assesses daily milk/weight patterns alongside a 7-day behavioral sequence and environmental THI context to enable real-time wellness triage.
4. To establish an Explainable Prescriptive Engine utilizing Grad-CAM to provide transparent and actionable non-medical management instructions to farmers.

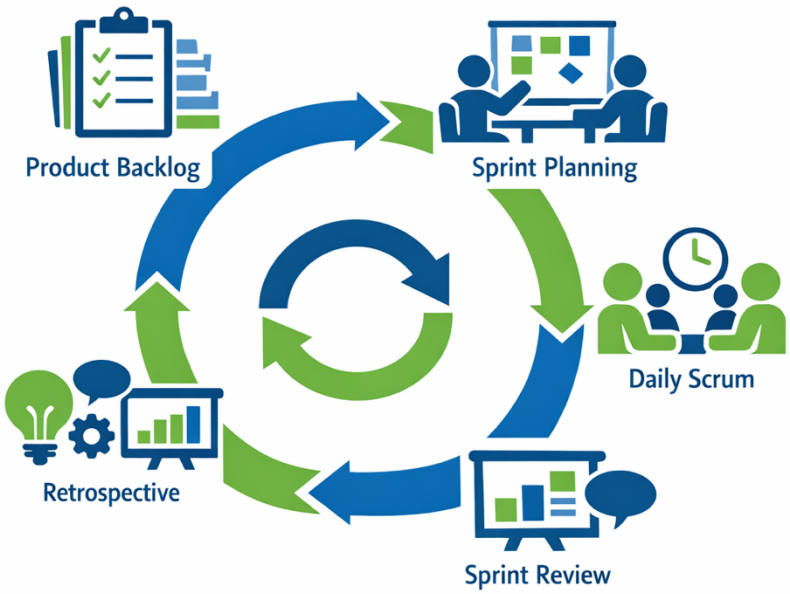
3. METHODOLOGY

3.1 Research Approach and SDLC

In this project, the research methodology will be Design Science Research (DSR), which aims to develop and evaluate an innovative IT artifact (the ADRS Triage Component) to address the real-world agricultural issue of delayed veterinary interventions [16].

To manage the software development life cycle within this DSR model, the Agile Scrum framework is applied [17]. Scrum is an iterative software development model that offers complete transparency and quick responsiveness. This is a highly appropriate method to use when working on Artificial Intelligence projects, where model accuracies and feature requirements dynamically change during the implementation phase.

Figure 1: Agile Scrum Methodology



3.2 Research Design / Framework

In order to make sure the AI-driven wellness artifact is developed in a rigorous way, the study adheres to the Design Science Research (DSR) cycles. The structure comprises of the steps to be followed:

Step 1: Problem Identification: The identification of the losses incurred in the dairy industry by using the reactive healthcare models.

Step 2: Objectives of the Solution: The development of multi-modal fusion and 2D vision requirements to narrow the triage gap.

Step 3: Design and Development: The design of the ADRS Triage system in terms of architecture will take place during this stage. The artifact is created like a continuous monitoring device that assesses daily records of milk weight and body weight in order to identify any statistical deviations . This continuous observation is a key initiator, as well as a human user override, to start a 7-day deep diagnostic triage window. The research aims at developing a late-fusion neural network that has the ability to combine multi-modal inputs such as daily logs and 7 days behavioral sequences to GPS-based environmental THI data to automated the cattle wellness classification procedure.

Step 4: Demonstration and Evaluation Testing the directly integrated system with manual veterinary benchmarks and visual-only baselines.

3.3 Feasibility Study

The viability of the proposed ADRS Triage system and its practical implementation was analyzed:

- **Technical Feasibility:** The project demands advanced knowledge of Python, FastAPI, YOLOv8 computer vision pipelines, in addition to Late-Fusion neural networks. Microsoft Azure Cloud GPU instances are available, which implies that the technical capability to process multi-modal data is present.
- **Economic Feasibility:** The system operates using low-cost 2D mobile cameras and open-source AI models. Moreover, the system is highly economically feasible because utilizing university-provided student Azure credits will fully cover the expenses of cloud hosting.
- **Operational Feasibility:** The system provides an on-demand triage tool, which is ideally compatible with the operational needs of farmers to monitor the health of cattle quicker, without the requirement of a continuous 24/7 monitoring infrastructure.

3.4 Data Collection and Ethical Considerations

- **Data Collection Strategy:** The key data that will be used in this study is the two-dimensional photographs of high milk content foreign cattle breeds including Jersey and Holstein-Friesian taken in small-scale tropical dairy farms. Data collection will be carried out in the two-level systematic manner: First, the data set consists of daily persistent data on milk production and body weights-scale data, which are registered by the farmer [18]. This is complemented by a period of 7 days of intensive behavioral record including water intake, feed intake, which is either automatically activated by the anomaly detection of the system or by the farmer [19]. At the same time, environmental information, such as ambient temperature and humidity in real-time are dynamically obtained through the Weather APIs with the help of GPS to compute Temperature-Humidity Index (THI) [20]. This multi-modal methodology will make

sure that AI model takes into consideration both physiological trends and environmental factors of the tropics which are stressors.

- **Ethical Considerations:** The farm owners of the involved farms will be given all the details about the study and will be obligated to sign a written consent form on all the images and farm data collected. All data obtained will be entirely anonymized so as to ensure that the privacy of the participants is not compromised. Moreover, the data collection is also a purely observational and non-invasive procedure, that is, the animals are not subjected to any physical or psychological violation, which complies with the general ethics of animal welfare.

3.5 Technical Methods and Algorithms

The central processing pipeline consists of four major steps that combine computer vision, environmental context, and physiological measurements in a well-integrated manner:

- **Stage 1: Advanced Image Processing (BCS Engine):** Raw mobile photographs are processed with Contrast-Limited Adaptive Histogram Equalization (CLAHE) to enhance skeletal shadows, and Gaussian Filtering is used to purge environmental noise. A YOLOv8 model isolates the rear-quarter view of the animal from complex backgrounds [18]. Afterwards, important anatomical keypoints (Hooks, Pins, Spine) are determined using a custom CNN-based Regression model to compute an objective Body Condition Score (BCS).
- **Stage 2: Environmental Context Integration:** The mobile device uses GPS positioning to request external Weather APIs to gather real-time temperature and humidity. It mathematically computes the Temperature-Humidity Index (THI) to account for tropical heat stress [19].
- **Stage 3: Multi-modal Data Fusion:** A Late-Fusion Neural Network mathematically combines the Visual Feature vector (BCS), the Physiological Sequence (7-day logs), and the Environmental Feature vector (THI) to produce a single Wellness Score classification (Healthy, At Risk, or Unhealthy) [20].
- **Stage 4: Explainable AI (XAI):** The system uses Grad-CAM algorithms to create visual heatmaps over the uploaded photo [21]. This highlights the particular anatomical areas that the AI paid attention to, and then a prescriptive logic engine is used to map the anomalies into actionable, non-medical management protocols.

3.6 Tools, Platforms and Alternatives Assessed

The technology stack was carefully chosen and based on several options after consideration to make sure that the system manages the multi-modal data processing effectively and can be easily accessible to the target population in accordance with the Software Engineering computing pillar.

Table 2 : Technical Stack, Justification, and Alternatives Evaluated

Tool / Platform	**Category**	**Alternatives Evaluated**	**Justification for Selection**
Python, PyTorch	Deep Learning	C++, Java, TensorFlow	Python was chosen because it provides an abundance of community resources, tutorials, and pre-trained models, drastically reducing development time. PyTorch was selected over TensorFlow due to its dynamic computational graph, providing higher flexibility for custom Late-Fusion.
YOLOv8 & OpenCV	Computer Vision	Faster R-CNN, SSD	Faster R-CNN was rejected due to high inference latency on mobile-captured images. YOLOv8 was selected for its optimal balance of real-time speed and boundary box accuracy.
Microsoft Azure & FastAPI	Backend Server & Cloud	AWS, GCP, Node.js	Microsoft Azure was specifically selected because the university student email provides free cloud credits, ensuring zero hosting costs for GPU instances. Node.js lacks native AI support. FastAPI is lightweight, asynchronous, and optimized for Python.
React.js	Frontend Development	React Native, Flutter, Native Android	Selected to develop a Responsive Web Application instead of a native mobile app. This architectural decision was made because rural farmers often lack the technical literacy to regularly download and install app updates from app stores. A responsive web approach guarantees they always access the latest version automatically via their mobile browser. Furthermore, a web application allows for a seamless, fully automated CI/CD deployment pipeline without the rigorous and delayed review processes of mobile app stores.

3.7 High-Fidelity User Interfaces

Modern and friendly user interfaces will be used to make sure that the usability of the tool is high among the rural farmers who are not very technological. The UI focuses on visual feedback rather than sophisticated text input and is designed with a mobile-first responsive web approach.

Figure 2: High-Fidelity UI Wireframes

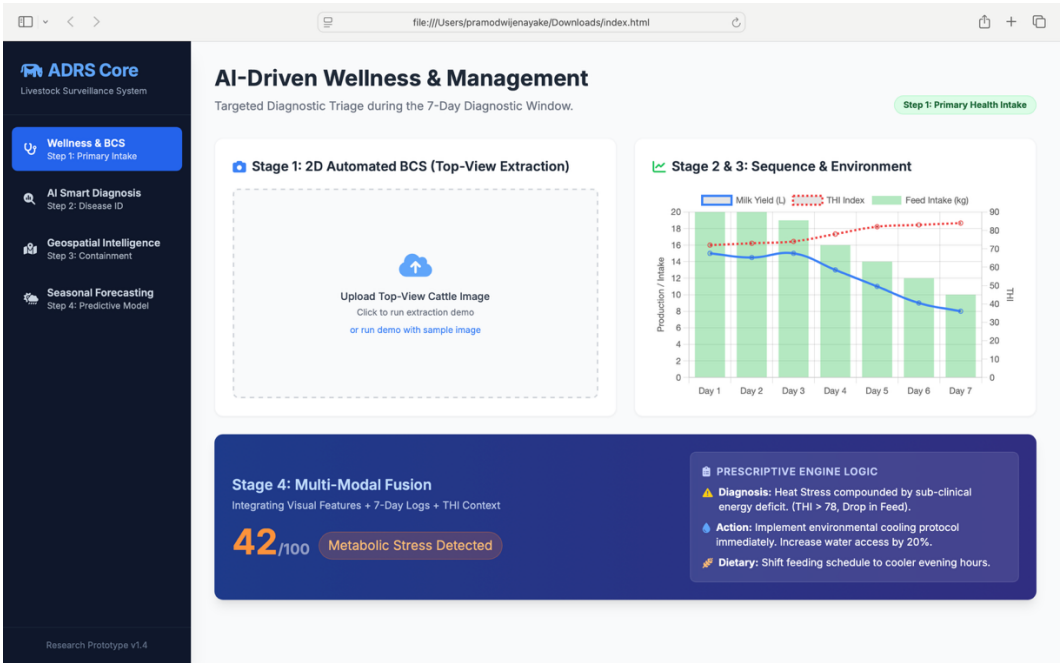
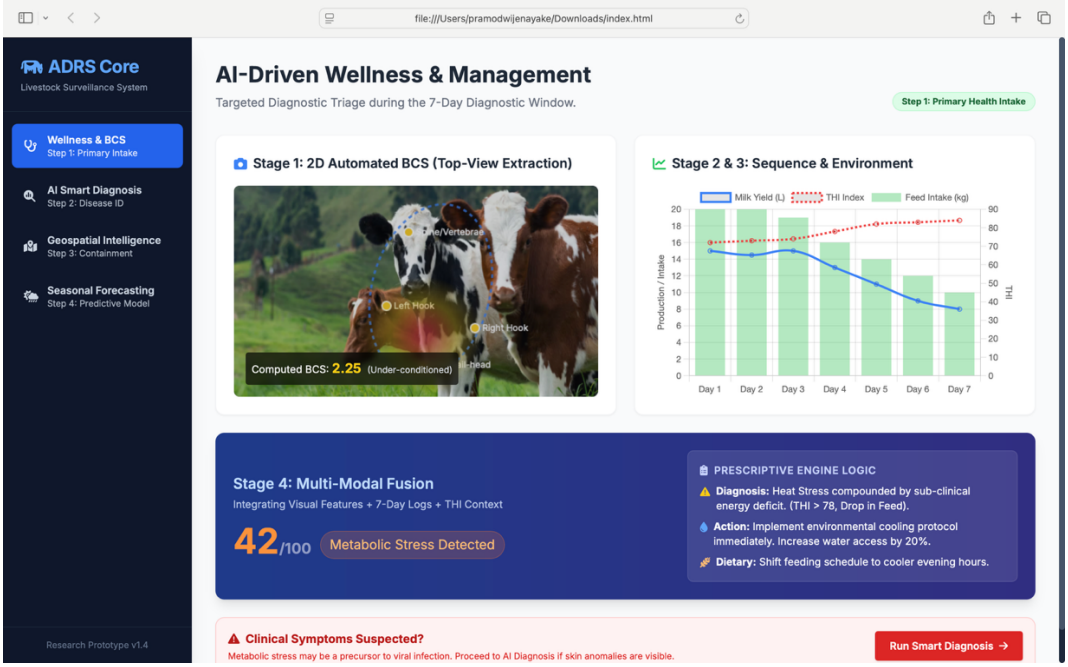


Figure 3: High-Fidelity UI Wireframes



3.8 Validation Strategy and Baselines

Strict performance evaluation of the system against laid-down baselines using quantitative measures of industry standards will be carried out to verify reliability before the system is implemented:

- **Baseline Comparison:** The proposed multi-modal Late-Fusion model will be directly measured against a **Baseline Visual-Only Model** (a standard CNN with neither 7-day logs nor THI context). This method mathematically demonstrates the performance improvement achieved by adding environmental and physiological data.
- **Vision Pipeline Metrics:** To determine the effectiveness of the bodily segmentation of the cow, **Intersection over Union (IoU)** and **Mean Average Precision (mAP)** are utilized.
- **BCS Estimation (CNN):** Assessed by **Root Mean Square Error (RMSE)** and **Mean Absolute Error (MAE)** based on the differences between the AI-computed BCS values and the manual BCS values provided by expert veterinary officers.
- **Triage Classification Metrics:** The overall diagnostic triage ability will be assessed using a **Confusion Matrix**, which will compute the total **Accuracy, Precision, Recall, and F1-Score** of the system in comparison to the baseline.

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4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and its Components

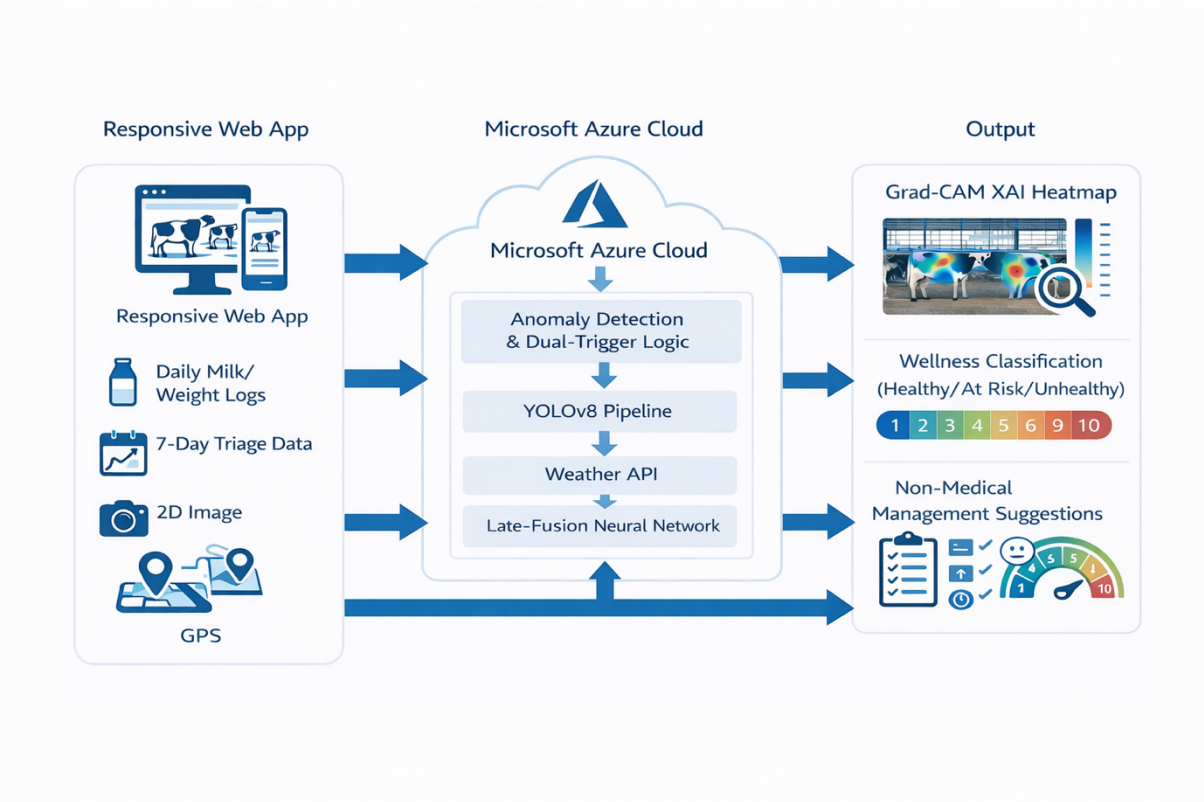
The proposed ADRS Triage Component is a state-of-the-art smart agricultural platform that offers real-time and objective cattle health assessments in a cloud-based environment. The system transforms rural dairy farming from a reactive healthcare model to a proactive one. It is built upon three major architectural layers:

1. **Client Presentation Layer:** A responsive web application developed using React.js. This enables farmers to use the system seamlessly through mobile browsers without the need to frequently update native apps.
2. **Backend API & Routing Layer:** An asynchronous, high-performance FastAPI server running on Microsoft Azure to securely aggregate and preprocess data.
3. **AI Processing Engine:** The core research layer that executes multi-modal deep learning inferences on cloud GPUs.

4.2 System Diagram of Components

Figure 4.1 represents the structural and logical layout of the high-level architecture, showing the clear demarcation between the client-side presentation layer and the cloud-based processing layers.

Figure 4: High-Level System Architecture Diagram



4.3 Highlight of the Individual Component and its Parts

In spite of the fact that the general system deals with data routing and UI rendering, this study focuses on the AI Processing Engine only. It is the critical research element and includes 5 specialized sections to aid the dual-trigger triage logic:

1. **Persistent Monitoring Unit:** This is a logic module that is always looking at the daily milk and weight records in order to identify any statistical outliers as a triggering point in autonomic operation.
2. **The Vision Pipeline:** YOLOv8 is used to conduct an anatomical breakdown of the rear-quarter of the cow and a CNN-regression model to derive an objective Body Condition Score (BCS) based on 2D images.
3. **The THI Calculator:** This is a context-aware unit, which retrieves ambient temperature and humidity through GPS-linked APIs and computes the Temperature-Humidity Index (THI) .
4. **The Late-Fusion Neural Network:** The mathematical combination of the visual BCS, the 7-day sequential behavioral logs with the THI vector into one classification of wellness.
5. **Explainable AI (XAI) Generator:** This uses the Grad-CAM algorithm to give visual heatmap, the areas of interest of the AI to the farmer.

4.4 Component Interactions and Data Flow Explanation

The ADRS Triage Component is an automated pipeline that takes in multi-modal data and transforms raw data into wellness insights that can be taken:

- **Phase 1**: Persistent Monitoring and Dual-Triggering: It begins with the farmer adding cattle profiles, and entering the daily milk and body weight scale-data in the React-based web application.
- **Phase 2**: Optimal Data Collection: After opening the triage window, the farmer is asked to give a 7-day history of water and feed consumption. At the same time, the GPS coordinates are automatically obtained on the phone to request outside Weather APIs (OpenWeatherMap) to obtain real-time ambient temperature and humidity.
- **Phase 3**: Multi-modal Data Preprocessing: The 2D image, which is captured at the last day of the triage, is exposed to the Contrast-Limited Adaptive Histogram Equalization (CLAHE) to boost skeletal characteristics to extract BCS. Meanwhile, behavioral logs will be turned into an ordered numerical amount and climate will transform into a Temperature-Humidity Index (THI) amount.
- **Phase 4**: Late-Fusion AI Inference: These multi-modal vectors are fused in a Late-Fusion Neural Network. This combined context is used in the model to come up with a final wellness category of Healthy, At Risk or Unhealthy.
- **Phase 5**: Open Result Delivery: The last category, including a Grad-CAM heatmap that provides visual explainability, is delivered to the web dashboard of the farmer through RESTful APIs. Longitudinal analysis All multi-modal records and diagnostic events are continuously stored in Azure Cosmos DB.

4.5 Scalability and Security Considerations

- **Scalability:** Image processing and neural network inferences are computationally dense. Therefore, the backend AI Processing Engine is entirely decoupled from the frontend web application. The FastAPI server on Microsoft Azure enables the AI engine to scale vertically (allocating more GPU instances) during peak usage periods, such as morning milking times, without throttling the user interface. Moreover, the use of Azure Cosmos DB (a NoSQL database) guarantees horizontal scalability for managing large volumes of unstructured diagnostic records.
- **Security:** Agricultural data privacy is highly prioritized. To avoid man-in-the-middle attacks, all data transmitted between the React web application and the FastAPI server is encrypted using HTTPS (TLS 1.2/1.3). Stateless JSON Web Tokens (JWT) are used for authentication and API route protection. Most importantly, all uploaded 2D images and diagnostic logs are completely anonymized—stripped of any personal farmer identification—before being stored in the database, fully aligning with ethical data privacy standards.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Identifying the stakeholders is a crucial step in the requirement engineering process. The ADRS Triage Component is a successful implementation that is facilitated using the following key stakeholders:

- **Primary Users (Small-scale Dairy Farmers):** Access the responsive web app to upload cattle images and behavioral logs to receive instant wellness scores.
- **Veterinary Officers and Technicians:** Utilize objective BCS data and Grad-CAM heatmaps to aid in clinical diagnosis and interventions.
- **Regional Dairy Cooperatives:** Oversee aggregated data regarding herd wellness to ensure the production of milk remains at a steady level in the area.
- **System Administrators:** In charge of the administration of the Microsoft Azure cloud computing infrastructure and the maintenance of AI model performance.

5.2 Functional Requirements

The functional requirements provide the main steps which the system is supposed to take so as to accomplish the objectives of the research. These are detailed in the table below:

Table 3: Functional Requirements for ADRS Triage Component

Requirement ID	Requirement Name	Description	Priority
FR-01	User Authentication	The system shall allow farmers to securely register and login to the ADRS web dashboard.	High
FR-02	Persistent Daily Logging	The system shall allow farmers to record daily metrics for milk yield and body weight scale-data to maintain a longitudinal health baseline.	High
FR-03	Dual-Trigger Logic	The system shall initiate a 7-day Deep Diagnostic window autonomously via anomaly detection in daily logs or manually via farmer observation.	High
FR-04	Triage Data Input	Once triggered, the system shall allow the input of a 7-day behavioral sequence covering feed and water consumption.	High
FR-05	Environmental Fetching	The system shall automatically retrieve GPS-linked ambient temperature and humidity to calculate the Temperature-Humidity Index (THI).	High
FR-06	AI BCS Estimation	The system shall utilize YOLOv8 for anatomical segmentation and CNN-regression for objective BCS computation from 2D images.	High
FR-07	Late-Fusion Classification	The system shall integrate image features, behavioral logs, and THI context to classify status as Healthy, At Risk, or Unhealthy.	High
FR-08	Explainable AI (XAI)	The system shall generate visual Grad-CAM heatmaps to interpret the AI model's focus areas for the farmer.	Medium
FR-09	Diagnostic History	The system shall persistently store all multi-modal records and diagnostic events in Azure Cosmos DB.	Medium
FR-10	Management Guidance	The system shall provide actionable, non-medical prescriptive suggestions based on the final triage result.	Medium

5.3 Non-Functional Requirements

The quality and performance of the ADRS Triage system is ensured using the following non-functional requirements:

- **Usability:** The interface should be minimalist and icon-driven to accommodate farmers who are not technologically savvy.
- **Latency:** The AI engine must have the capacity to deliver a complete wellness triage solution within 10-15 seconds.
- **Security:** All data transfers should be encrypted over HTTPS, and diagnostic logs should be anonymized before storage to maintain the privacy of farmers.
- **Scalability:** The Azure cloud backend should be capable of scaling vertically to handle high compute demands during peak milking times.
- **Availability:** The system is expected to maintain 99.5% uptime for on-demand diagnostic availability.

6. COMMERCIALIZATION PLAN

6.1 Customer Persona and Target Market

The ADRS Triage Component is primarily aimed at the basis of the dairy economy in Sri Lanka:

- **Primary Target:** Jersey and Holstein-Friesian small-scale dairy farmers in rural Sri Lanka who are currently restricted by a reactive healthcare paradigm .
- **Customer Persona:** A farmer with a smartphone that has no immediate or affordable access to professional Body Condition Scoring (BCS) and thermal stress diagnostics.
- **Secondary Market (Veterinary Surgeons and Technicians):** The system enables prioritizing emergency cases and providing remote consultations to private and governmental veterinarians with the help of pre-clinical alerts and Grad-CAM heatmaps provided by the system.
- **Final Market:** Dairy co-operatives and milk collection centers in the region that aim to stabilise the supply chain with the help of the aggregate herd wellness monitoring.

6.2 Value Proposition

The system provides a dedicated value, bridging the so-called Triage Gap between the current Precision Livestock Farming (PLF) models :

- **No Infrastructure Expense:** ADRS is supported by standard 2D mobile devices as opposed to the existing PLF solutions that use expensive 3D cameras and sensors .
- **Proactive Diagnosis:** It allows one to identify sub-clinical changes earlier before they develop into metabolic disorders or herd infections.
- **Objective Accuracy:** Substitutes subjective manual scoring with a multi-modal AI combination of BCS, 7-day behavioral logs, and real-time THI .

6.3 Revenue Model

A Freemium and Tiered Subscription model is proposed to achieve sustainability and extensive acceptance:

- **Freemium Tier:** The basic health classification (Healthy/Unhealthy) will be free in order to create a large user base.
- **Pay-Per-Report:** A micro-payment (e.g., LKR 150) to get a detailed diagnostic report with Grad-CAM heatmaps and management prescriptions.
- **B2B Licensing:** Charging dairy cooperatives and veterinary clinics for the use of a centralized dashboard to track cattle wellness scores within their network.

6.4 Cost Estimation

- **Cloud Infrastructure:** Monthly fees for Microsoft Azure GPU instances used for AI inference and Cosmos DB for data storage .
- **Model Refinement:** Costs of retraining the **YOLOv8 [18]** and **Late-Fusion models [20]** using new tropical cattle images.

- **Operation Costs:** Low API maintenance costs (Weather APIs) and community-based marketing.

6.5 Pricing Strategy

A Value-Based Pricing policy is used. Since a single clinical case could lead to a 20-30% loss of milk production, the diagnostic cost is a small preventive measure which would save the farmer thousands of rupees on veterinary bills and lost revenue.

6.6 Competitive Advantage

This system is based on real-life contextual information that global products tend to ignore :

- **Contextual Fusion:** This is the single system that matches 7-day logs to real-time **GPS-based THI [19]**.
- **Explainability:** Building user trust by using **Grad-CAM [21]** to show the exact areas of the anatomy that the AI used for identification

Table 4: Comparison of ADRS with Existing Alternatives

Feature / Criteria	Manual Veterinary Scoring	Industrial PLF Systems	ADRS Triage (Proposed)
Initial Hardware Cost	None (Service fee only)	Very High (LKR 500,000+)	Zero (Uses existing Smartphone)
Accessibility	Limited (Vet's availability)	High (But needs tech infra)	On-Demand (Access via Web)
Data Modalities	Manual observation only	Continuous IoT/3D Sensors	Multi-modal (Image+Logs+THI)
Early Warning	Low (Mostly reactive)	Very High	High (Proactive detection)
Environmental Context	Subjective	Integrated (High-cost)	Integrated (Free via GPS API)
Explainability (XAI)	Based on Vet's word	No (Black-box algorithm)	Yes (Grad-CAM Heatmaps)

6.7 Intellectual Property (IP) Considerations

- **Copyright Protection:** The FastAPI integration and Late-Fusion architecture custom code will be copyrighted.
- **Trade Secret:** The special mathematical weights and feature concatenation logic that were optimized to work with Sri Lankan cattle will be a trade secret.
- **Trademark:** The product will be registered as an official trademark under the name and logo of ADRS to avoid imitation.

7. BUDGET AND JUSTIFICATION

The ADRS Triage Component budget is strategically planned in a way that it reduces the reliance on hardware since it will utilize cloud-computed AI and current mobile technology. The approximate costs and justifications are shown in the following table:

Table 5:Project Budget and Justification

Category	Item Description	Estimated Cost (LKR)	Justification
Hardware	Mid-range Smartphone (Testing)	65,000.00	Required to test the responsive web application and 2D image capture quality in field conditions.
Hardware	Portable SSD (1TB)	45,000.00	Necessary for secure offline backup and transport of high-resolution bovine datasets collected in the field .
Travel	Traveling Cost (Data Collection & Testing)	20,000.00	Required for multiple field visits to rural farms for primary data collection and final system validation with veterinarians.
Cloud Services	Microsoft Azure GPU Instances	0.00 (Standard: ~45,000/mo)	Utilizing University Student Azure Credits to host YOLOv8 and Late-Fusion models for free.
Cloud Services	Encrypted Data Storage	15,000.00 (Total for pilot)	To facilitate compliance with image databases that generate OOD checks and diagnostic history.
Software	Google Colab Pro Subscription	19,200.00 (6 months)	To gain access to premium GPUs (NVIDIA T4/V100) for faster model iterations during training .
Human Effort	Research & Development	450,000.00 (Estimated)	Based on the industry average for a junior AI developer's effort over the 6-month project duration.
Total Outlay		614,200.00	

7.1 Justification for Expenses

- **Hardware Efficiency:** The budget does not necessarily require any costly 3D cameras or specialized IoT sensors, but a regular smartphone, which reveals the accessibility of the system by small-scale farmers.
- **Strategic Application of Cloud Credits:** With the help of the student program of Microsoft Azure, the project will remove the largest financial obstacle, i.e., high-performance GPU hosting.
- **Open-Source Leverage:** The use of Python and Python libraries will mean that no repeated software licensing fees are paid, and the cost of operating is set close to zero.
- **Human Capital:** The high-value research and complex multi-modal fusion logic specific to the Sri Lankan market is reflected in the estimated human effort cost.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown by member

Component 1: AI-Powered Smart Diagnosis (Computer Vision and Prognostics)

- Member 1: A.L.M Athulathmudali
- Key Tasks:

1. Collect and Annotate Primary and Secondary Data Sets for Local Cattle Breeds (i.e. Cattle Breeds found locally) for multiple concurrent diseases and their associated vitality indicators.
2. Design and Build Input Integrity Gate using YOLOv10-tiny and Out-of-Distribution (OOD) Additional Information.
3. Train Hierarchical Multi-Label Classifier (Swin Transformer) and Severity Quantification Module (Mask R-CNN).
4. Implement the Chain of Thought (CoT) transparent framework and prognostic survival engine.
5. Validate the diagnostic threshold using K-Fold cross-validation to ensure it has an 85% accuracy rate.

Component 2: Real-Time Geospatial Clustering & Trajectory Mapping

- Member 2: G.A Sandaru
- Key Tasks:

1. Gather and process both geographical and historical outbreak data regarding Sri Lanka's livestock.
2. Create density-based spatial clustering algorithms to detect contemporary (real-time) disease hot-spots.
3. Utilize physics-based spread modeling to determine both the localized outbreak velocities and trajectories.
4. Build an Interactive Intelligence Map for use by Government and Veterinary Agencies.
5. Build API endpoints to securely accept diagnostic payloads (type of disease and its severity) from the Computer Vision component.

Component 3: Seasonal Forecasting for Risk assessment

- Member 3: S.S Kumarasinghe
- Key Tasks:

1. Combine historical climate records, monsoon characteristics, and environmental variables related to tropical livestock diseases into a comprehensive data set.
2. Design non-linear machine learning forecast models (LSTM's or Conformal Inference) to enable effective temporal forecasting and the identification of sudden spikes in disease occurrence.
3. Create a local, seasonal "Risk Index" by combining relevant weather variables to designate the level of risk within select geographic areas.
4. Confirm the validity of the forecast model's precision when correlating the forecasted outbreak occurrences with recorded outbreak data.
5. Integrate the results of the predictive risk assessment with the central ADRS database to notify vulnerable farming areas.

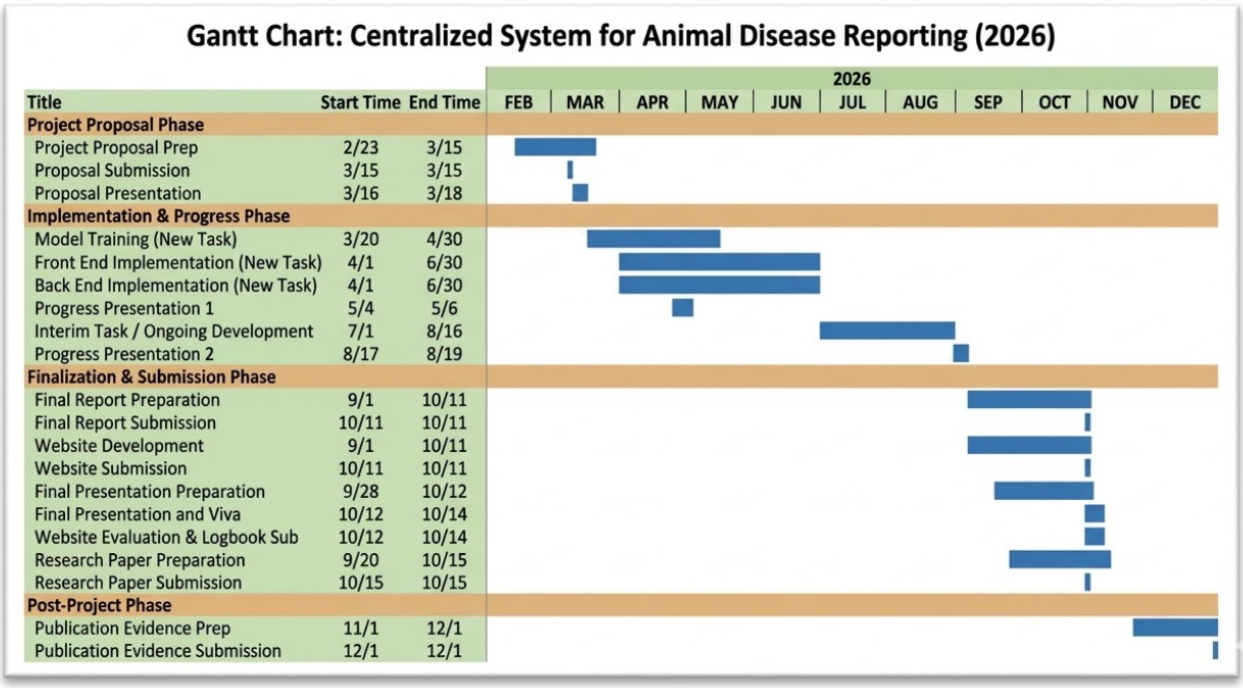
Component 4: Animal Health Detection (Wellness and Management)

- Member 4: W.M.P.J Wijenayake
- Key Tasks:

1. Develop and establish corporate standard Multiple Base Parameters for the recognition of physiological distress and physical wellness in local dairy cattle.
2. Create a 2D vision-based algorithm to automate the reading of Body Condition Scores (BCS) by using skeletal keypoint detection software.
3. Use the clean segmented masks of bovine animals created using algorithms from the Computer Vision Component to remove lesions when calculating the BCS.
4. Create a farmer view of the wellness of the animal that includes recording the physiological conditions and energy status of the herd.
5. Conduct field tests to compare automated reading of BCS to manually recorded readings prepared by veterinarians.

9. GANTT CHART

Figure 5: Gantt chart



Title	Start Time	End Time
Project Proposal Phase		
Project Proposal Prep	2/23	3/15
Proposal Submission	3/15	3/15
Proposal Presentation	3/16	3/18
Implementation & Progress Phase		
Model Training (New Task)	3/20	4/30
Front End Implementation (New Task)	4/1	6/30
Back End Implementation (New Task)	4/1	6/30
Progress Presentation 1	5/4	5/6
Interim Task / Ongoing Development	7/1	8/16
Progress Presentation 2	8/17	8/19
Finalization & Submission Phase		
Final Report Preparation	9/1	10/11
Final Report Submission	10/11	10/11
Website Development	9/1	10/11
Website Submission	10/11	10/11
Final Presentation Preparation	9/28	10/12
Final Presentation and Viva	10/12	10/14
Website Evaluation & Logbook Sub	10/12	10/14
Research Paper Preparation	9/20	10/15
Research Paper Submission	10/15	10/15
Post-Project Phase		
Publication Evidence Prep	11/1	12/1
Publication Evidence Submission	12/1	12/1

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11. APPENDICES

11.1 Project Risk Analysis

Table 6: Project Risk Analysis

Potential Risk	Severity	Mitigation Strategy
Poor image quality due to low light	High	Implementing CLAHE and Gaussian filtering for image enhancement.
Inaccurate logs provided by farmers	Medium	Using multi-modal Late-Fusion to cross-verify visual data with behavioral trends.
Connectivity issues in rural farms	Low	Designing the web app to support basic offline data logging capabilities.
User Adoption: Farmers' low tech literacy	Medium	Designing an icon-driven, minimalist UI for easy navigation.